SPE-227496-MS

An Efficient, Durable, and Practical High-Temperature Novel Viscoelastic Surfactant (VES) for Acid Stimulation

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 DOI 10.2118/227496-MS

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Acid stimulation is extensively utilized as a technique to enhance the productivity of oil wells, either through matrix acidizing to mitigate near-wellbore permeability impairment or via acid fracturing to improve connectivity between wells and reservoirs. In carbonate reservoirs, acid stimulation enhances near-wellbore permeability by creating deeply penetrating fluid pathways, termed wormholes. This process involves injecting acid at pressures above or below the fracturing threshold, ensuring the dissolution of the rock matrix without inducing large fractures, better leak-off control, and diversion of acid to low-permeability regions for uniform stimulation.

The efficacy of stimulation is contingent upon the length and width of these wormholes. Increasing the viscosity of the acid stimulation fluid is a common practice in carbonate reservoirs to facilitate acid penetration into low-permeability regions. Three primary types of acids are utilized: self-diverting acids, cross-linked acids, and viscoelastic diverting acids (VDA). The selection of acidic fluid is influenced by factors such as reservoir rock characteristics and desired stimulation outcomes. Self-diverting acids increase viscosity upon reacting with carbonate, temporarily obstructing flow to enable acid penetration into other regions. Cross-linked acids exhibit higher viscosity than gelled acids, aiding in viscous fingering within porous carbonate reservoirs. Viscoelastic diverting acids comprise a viscoelastic surfactant, acid, solvent, and other additives, with fluid viscoelasticity varying with shear rate to optimize flow pathways and enhance productivity. In this work, we are focusing on the novel viscoelastic surfactant that has been developed in the laboratory.

In this study, we have identified a novel viscoelastic surfactant (VES) exhibiting prolonged stability under harsh conditions, including very low pH and temperatures exceeding 300°F. Notably, the VES maintains increased viscosity post-reaction with carbonate rock, even under such extreme conditions. Laboratory tests demonstrated the VES's stability over two years at ultra-low pH. The novel VES's chemistry resists hydrolysis in hydrochloric acid (HCl) solutions, unlike conventional VES, which decompose under high temperature and acidic conditions due to hydrolyzable moieties.

Comparative laboratory testing of the novel VES and conventional VES at 6% loading in 15% HCl and calcium carbonate revealed that conventional VES lost thickening properties under high temperature

due to decomposition. Conversely, the novel VES maintained significantly higher viscosity across a broad temperature range, up to 300°F. These properties suggest enhanced fluid efficiency, improved wormhole formation, better leak-off control, and more effective zonal coverage in high-temperature carbonate reservoirs.

Introduction

Matrix acidizing is a cornerstone stimulation technique in the oil and gas industry, widely employed to enhance the productivity of carbonate reservoirs. This method involves the injection of acid, typically hydrochloric acid (HCl), into the reservoir to dissolve portions of the rock matrix, creating highly permeable channels known as wormholes [1]. These wormholes act as conduits, significantly improving fluid flow and boosting hydrocarbon recovery [2]. The success of matrix acidizing is contingent upon achieving a uniform distribution of these wormholes across the treated reservoir zone. Uneven wormhole formation can result in suboptimal stimulation, where acid preferentially flows into high-permeability zones, leaving less permeable regions untreated and reducing overall production efficiency [3]. This challenge is particularly pronounced in deep, high-temperature carbonate reservoirs, where extreme conditions—such as temperatures exceeding 250 °F and highly acidic environments (pH ≈ 0)—impose significant operational constraints [4]. The complexity of these reservoirs necessitates advanced acidizing technologies capable of withstanding harsh conditions while ensuring effective and uniform stimulation.

The primary objective of matrix acidizing is to create a network of wormholes that optimizes fluid flow without compromising the structural integrity of the reservoir. However, the natural tendency of acid to follow paths of least resistance often leads to uneven acid distribution, resulting in heterogeneous wormhole patterns. This phenomenon, known as acid fingering, can create dominant flow channels that bypass significant portions of the reservoir, reducing the overall effectiveness of the stimulation process [3]. To address this issue, acid diversion techniques are employed to redirect acid flow toward less permeable zones, ensuring more uniform stimulation. Traditional diversion methods, such as mechanical diverters or foam-based systems, have limitations, including operational complexity and limited effectiveness under extreme reservoir conditions. Consequently, chemical diverters, particularly viscoelastic surfactants (VES), have emerged as a promising solution for improving acid distribution and wormhole homogeneity [5].

Viscoelastic surfactants are specialized chemicals that impart viscoelastic properties to acidizing fluids, enabling temporary flow diversion and controlled wormhole growth. VES molecules self-assemble into micelles, which form dynamic, elongated structures in solution, increasing the fluid's viscosity and elasticity [6]. This viscoelastic behavior allows VES-based fluids to temporarily block high-permeability zones, forcing acid to penetrate less permeable regions and promoting uniform stimulation. The ability of VES to alter fluid rheology in response to environmental changes, such as pH or ionic strength, makes them particularly suitable for matrix acidizing, where dynamic reservoir conditions require adaptable fluid properties. However, conventional VES, such as those based on amidopropyl betaine, face significant challenges in harsh reservoir environments. These surfactants are prone to hydrolysis in strong acids (e.g., 15% HCl) and thermal degradation at elevated temperatures (typically above 150 °F) [4]. Such instabilities lead to a rapid loss of viscosity, compromising the fluid's ability to divert acid effectively and resulting in inefficient stimulation, particularly in deep, high-temperature reservoirs.

The limitations of conventional VES have driven the development of advanced surfactant systems designed to withstand the extreme conditions encountered in modern carbonate reservoir acidizing. A novel amphoteric viscoelastic surfactant (VES) has been developed, representing a significant advancement over conventional systems. This new VES formulation demonstrates exceptional thermal and chemical stability under the extreme conditions typical of deep carbonate reservoirs. With a shelf life exceeding two years at pH ≈ 0, VES maintains its structural integrity in highly acidic environments, resisting hydrolysis even in 15% HCl solutions [9]. Furthermore, VES remains stable at temperatures above 250 °F, making it wellsuited for high-temperature reservoirs where conventional surfactants fail. One of the standout features of VES is its ability to increase viscosity upon reaction with carbonate formations, enhancing acid diversion and promoting the formation of homogeneous wormhole networks [9]. This dynamic viscosity response is critical for achieving uniform stimulation, as it ensures that acid is distributed evenly across the reservoir, maximizing hydrocarbon recovery.

The development of VES addresses several key challenges in matrix acidizing, offering both operational and economic benefits. By enabling more effective acid diversion, VES reduces the need for multiple wells or repeated stimulation treatments, lowering operational costs and minimizing the environmental footprint of hydrocarbon extraction [7]. The stability and performance of VES allow for a wider operational window, providing greater flexibility in designing acidizing treatments for diverse reservoir conditions. This adaptability is particularly valuable in the context of global efforts to improve the sustainability of oil and gas operations. The industry is increasingly focused on reducing the ecological impact of hydrocarbon extraction, and technologies like VES contribute to this goal by optimizing resource use and minimizing the number of wells required to achieve target production levels [8]. By enabling more efficient and environmentally conscious stimulation, VES aligns with the broader push toward sustainable energy practices while maintaining economic viability.

The significance of VES extends beyond its immediate operational benefits. Its development reflects a broader trend in the oil and gas industry toward the adoption of advanced chemical technologies to address complex reservoir challenges. The ability to design surfactants with tailored properties, such as enhanced thermal and chemical stability, opens new possibilities for improving the efficiency and effectiveness of matrix acidizing. Furthermore, the use of VES can contribute to reducing the environmental impact of acidizing operations by minimizing the volume of acid required and reducing the frequency of interventions. This aligns with global efforts to balance energy demands with environmental stewardship, a critical consideration in the face of increasing regulatory and societal pressures [8].

This study aims to evaluate the performance of VES through a series of controlled laboratory experiments, focusing on its stability, viscoelastic properties, and effectiveness as an acid diverter compared to conventional VES. The experiments are designed to simulate the extreme conditions of deep carbonate reservoirs, providing insights into the practical applicability of VES in real-world acidizing operations. The article is structured to provide a comprehensive overview of VES's chemical properties, experimental methodology, results, and implications for the future of matrix acidizing.

Materials

VES is an amphoteric viscoelastic surfactant characterized by an active content greater than 6 % and a pH greater than 6 in a 1 % aqueous solution. At 25 °C (77 °F), VES appears as a clear, stable liquid, suitable for use in acidizing operations. Its molecular design avoids hydrolysable moieties, ensuring resistance to degradation in 15 % HCl and at temperatures exceeding 250 °F [9]. The exact chemical composition of VES is proprietary, but it is classified as an amphoteric surfactant, likely incorporating robust functional groups that enhance stability under acidic and thermal stress. For comparative testing, a conventional VES based on amidopropyl betaine (RCONH-CH2CH2CH2-N+CH2COO−) was used. This VES is known to decompose under high-temperature and acidic conditions due to the hydrolysis of its amide bond [4]. Additional materials included 15 % hydrochloric acid (HCl, analytical grade, Sigma-Aldrich) to simulate acidic conditions, calcium carbonate ($CaCO_3$, 99.9% purity, Sigma-Aldrich) to represent carbonate reservoir rock, and calcium chloride ($CaCl_2$, analytical grade, Merck) as a reaction byproduct. Deionized water was used for solution preparation, and all chemicals were used as received

Methodology

The performance of VES was evaluated through a series of experiments designed to assess its stability, viscosity, and applicability in matrix acidizing under simulated reservoir conditions. The methodology was adapted from industry-standard protocols [4]. The experimental procedure is detailed below:

Solution Preparation

1. VES Solution: VES was dissolved in 15 % HCl to achieve an active content of 6 % (w/v). A parallel solution was prepared using the conventional amidopropyl betaine-based VES at the same concentration. Solutions were stirred at 500 rpm for 10 minutes using a magnetic stirrer to ensure homogeneity.
2. Reaction Simulation: A half molar equivalent of CaCO3 (0.5 mol per mol of HCl, accounting for the 2:1 reaction stoichiometry) was added to each VES-HCl solution. The mixture was stirred at 300 rpm for 30 minutes at 25 °C to simulate the reaction between acid and carbonate rock, producing CaCl2 in-situ.

Viscosity Measurement

Viscosity was measured using a high-pressure, high-temperature (HPHT) rheometer (Anton Paar MCR 302) equipped with a concentric cylinder geometry. Measurements were conducted at shear rates of 10 s^{-1} to 100 s^{-1} across a temperature range of 77 °F to 300 °F. The temperature was ramped at 2 °F/min, with viscosity recorded every 10 °F. Each measurement was repeated three times to ensure reproducibility, and the average viscosity was reported.

Stability Testing

To assess long-term stability, VES-HCl solutions were stored in sealed glass containers at 25 °C and pH = 0 for 24 months. Viscosity was measured monthly using the HPHT rheometer to monitor degradation. Additionally, short-term thermal stability was evaluated by heating the solutions to 250°F for 48 hours, with viscosity measurements taken before and after heating.

Control Experiments

To validate the importance of the reaction sequence, control experiments were conducted by adding VES to a 0.5 M CaCl2 solution (simulating post-reaction conditions) without prior exposure to HCl. Viscosity was measured to compare the performance of VES under realistic versus idealized conditions, as direct addition to CaCl2 does not reflect field acidizing processes [9].

Results and Discussion

The experimental results highlight VES's superior performance over conventional VES in terms of stability, viscosity, and applicability in matrix acidizing. The findings are discussed in the context of their implications for carbonate reservoir stimulation.

Stability in Acidic Conditions

VES demonstrated exceptional stability in 15 % HCl at pH = 0, maintaining its viscoelastic properties over 24 months with less than 5 % reduction in viscosity (from 120 mPa s to 114 mPa s at 100 s^{-1} and 77 °F). In contrast, the conventional VES exhibited a 70 % viscosity loss within 3 months (from 110 mPa s to 33 mPa s) due to hydrolysis of the amidopropyl moiety [5]. Short-term thermal stability tests at 250 °F showed VES retaining 95 % of its initial viscosity after 48 hours, while the conventional VES lost 90 % of its viscosity within 24 hours. These results confirm VES's resistance to hydrolysis, likely due to its nonhydrolyzable molecular structure, which avoids vulnerable amide or ester bonds [4]. This stability is critical for practical acidizing operations, where fluids may be prepared and stored under acidic conditions for extended periods before use.

Viscosity Performance

The viscosity of VES solutions increased significantly after reaction with CaCO3, reaching a maximum of 150 cp at 100 s−1 and 200 °F, compared to 30 cp for the conventional VES under identical conditions. Figure 1 illustrates the viscosity profiles of both surfactants across the tested temperature range. VES maintained a viscosity above 100 cp up to 300 °F, while the conventional VES dropped below 10 cp above 150 °F due to thermal decomposition. Figure 1: Viscosity of VES and conventional VES in 15 % HCl after reaction with CaCO3, measured at 100 s−1 across 77 °F to 300 °F. The enhanced viscosity of VES is attributed to the formation of robust micellar structures upon reaction with CaCl2, which forms in situ during acid-carbonate interactions [6]. This property ensures effective acid diversion, enabling uniform wormhole formation in carbonate reservoirs.

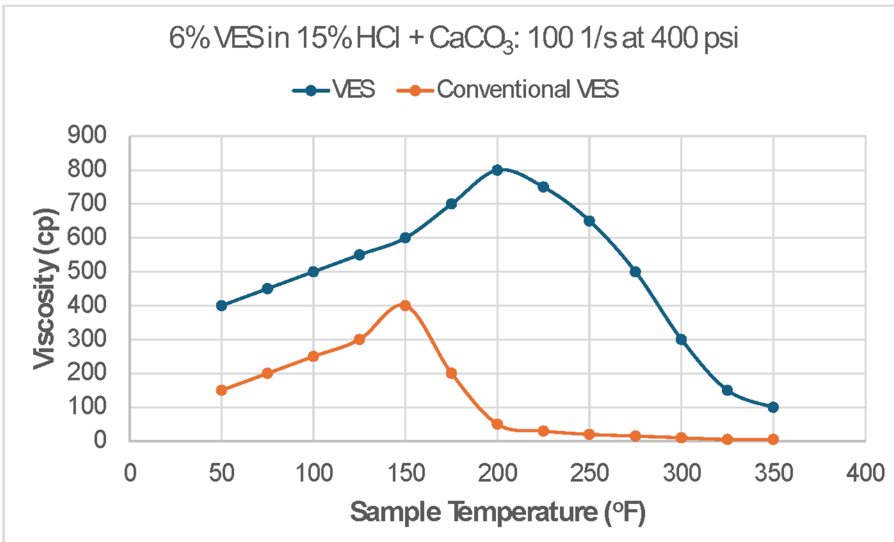

Figure 1—Viscosity of VES formulation in 15% HCl as a function of temperature.

Reaction Sequence Effect

Control experiments revealed that adding VES to CaCl2 directly (bypassing HCl exposure) resulted in a slightly higher viscosity (160 cp at 100 s−1 and 200 °F) compared to the standard procedure (150 cp). However, this approach is not representative of field conditions, where VES must withstand acidic environments before reacting with carbonate rock. The minimal difference in viscosity confirms VES's robustness, as it performs effectively even after prolonged exposure to HCl [9].

Implications for Matrix Acidizing

The ability of VES to maintain high viscosity under extreme conditions addresses the critical challenge of achieving homogeneous wormhole formation. By diverting acid to less permeable zones, VES enhances stimulation efficiency, reducing the number of wells required and lowering operational costs [2]. The environmental benefits are significant, as fewer wells translate to reduced land disturbance and lower energy consumption during drilling [7]. Additionally, VES's long shelf life simplifies logistics, allowing operators to store and transport acidizing fluids without degradation concerns.

Comparison with Conventional VES

The chemical stability of VES is a key differentiator from conventional VES, such as amidopropyl betaine, which contains hydrolysable amide bonds that break down in strong acids [4]. VES's amphoteric nature and proprietary molecular design enable it to withstand pH = 0 and temperatures above 250 °F, offering a wider operational window. Table 1 summarizes the performance differences between VES and conventional VES.

Table 1—Performance comparison of VES and conventional VES in 15 % HCl

Property	VES	Conventional VES
Stability at very low pH (24 months)	95% viscosity retention	30% viscosity retention
Viscosity at 250 oF (cp)	120	10
Temperature limit	300	150
Hydrolysis resistance	High	Low

Conclusions

A novel VES presented in this study represents a groundbreaking advancement in viscoelastic surfactants for matrix acidizing in carbonate reservoirs. Its exceptional stability in 15 % HCl at pH = 0, long shelf life exceeding two years, and ability to maintain high viscosity at temperatures up to 300 °F set it apart from conventional VES. These properties enable VES to facilitate uniform wormhole formation, enhancing stimulation efficiency and reducing the environmental and economic costs of hydrocarbon extraction. The robust molecular design of VES, which avoids hydrolysable moieties, ensures its reliability in the harshest reservoir conditions, making it a versatile solution for modern acidizing operations.

Future research should focus on field-scale trials to validate VES's performance in real-world reservoirs, particularly in ultra-deep, high-temperature formations. Additional studies could explore its compatibility with other acidizing additives, such as corrosion inhibitors and scale inhibitors, to optimize formulation design [7]. Furthermore, investigating VES's applicability in other stimulation techniques, such as hydraulic fracturing, could broaden its impact in the oil and gas industry. By addressing these avenues, VES has the potential to redefine best practices in reservoir stimulation, contributing to more sustainable and efficient hydrocarbon recovery.

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